10.1.2 . Reverse a String

A] Algorithm

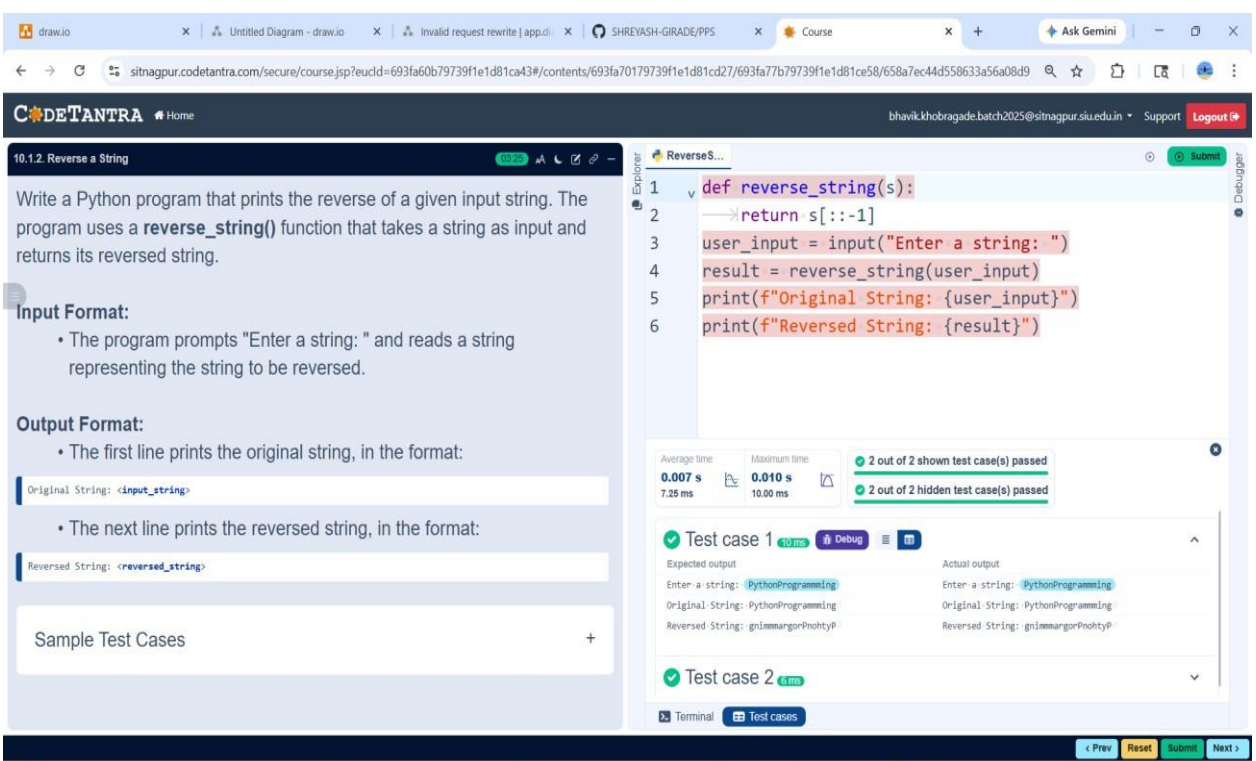
Step 1: Start
Step 2: Read input string from the user and store it in user_input
Step 3: Define a function reverse_string(input_string)
Step 4: Inside the function, initialize an empty string re = ""
Step 5: Find the length l of input_string
Step 6: Loop from index l-1 down to 0 (reverse order):
    Step 6a: Append the character at current index to re
Step 7: After the loop, return re (the reversed string)
Step 8: Call reverse_string(user_input) and store the result in result
Step 9: Print the original string user_input
Step 10: Print the reversed string result
Step 11: End

B] code

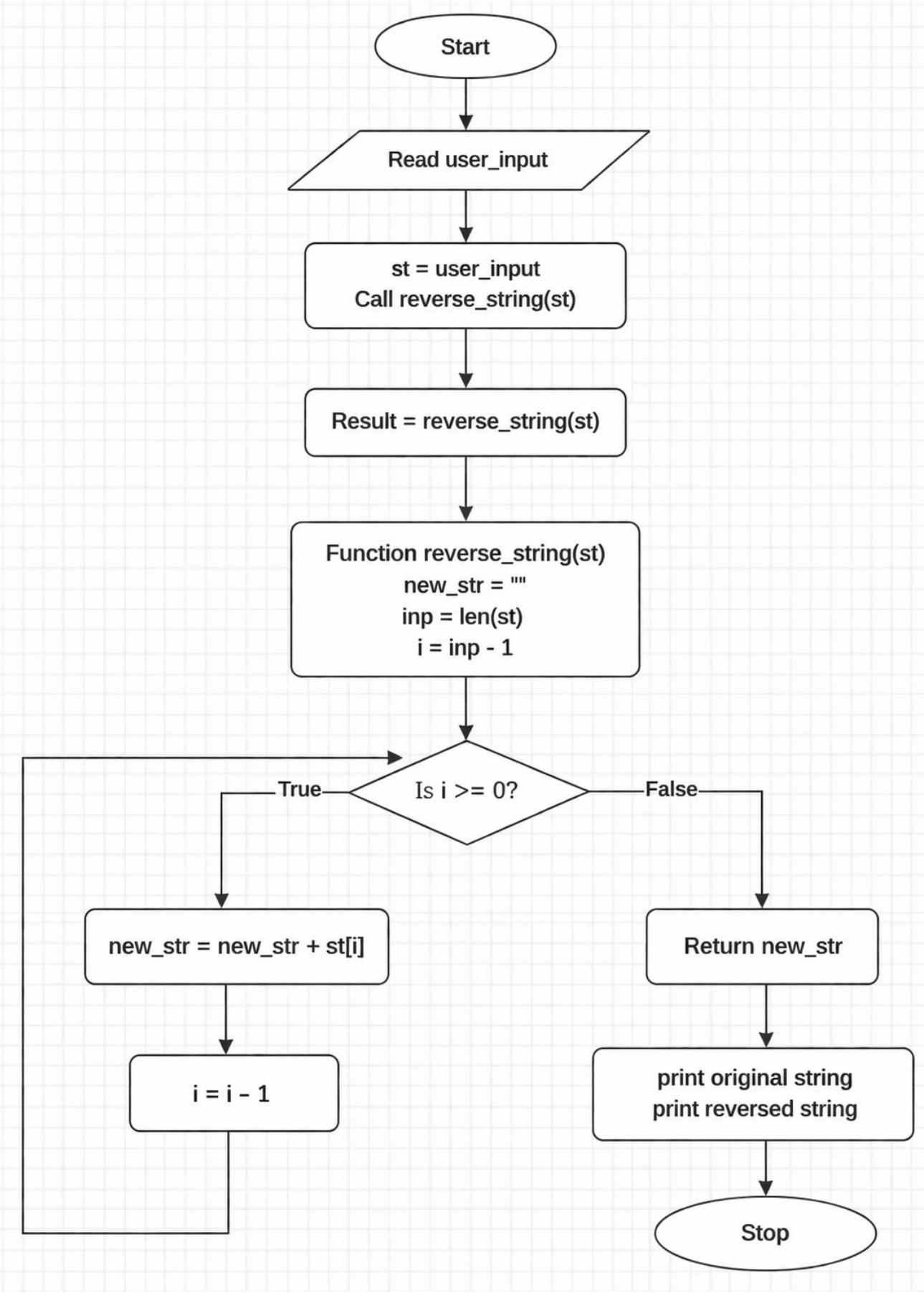
```
def reverse_string(input_string):
        re=""
        l = len(user_input)
        for i in range(l-1,-1,-1):
                re += user_input[i]
        return re


user_input = input("Enter a string: ")
result = reverse_string(user_input)
print(f"Original String: {user_input}")
print(f"Reversed String: {result}")
```

D] output



C] flowchart


Start
Read user_input
st = user_input
Call reverse_string(st)
Result = reverse_string(st)
Function reverse_string(st)
new_str = ""
inp = len(st)
i = inp - 1
Is i >= 0?
True
False
new_str = new_str + st[i]
i = i – 1
Return new_str
print original string
print reversed string
Stop